Dimension-5 ORBIT Comparison to a Sign-Flipped Einstein-Hilbert Reference

ORBIT

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1 Setup

This report records the final lightweight comparison run carried out after introducing an explicit user-controlled kinetic-sign convention into the ORBIT comparison layer. The comparison was intentionally restricted to mass dimension

$$d \leq 5$$

in order to avoid the substantially more expensive full $d \leq 6$ rerun.

The conventions used for this final run are:

- mostly-minus Lorentzian signature;
- metric perturbation $g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu}$;
- reference action
$$\mathcal{L}_{\mathrm{ref}} = \Lambda_{\mathrm{ref}}\sqrt{-g} - \frac{2}{\kappa_{\mathrm{ref}}^2}\sqrt{-g}\,R;$$
- canonical normalization of the Matchete input to the predetermined quadratic target sign
$$s_{\mathrm{kin}} = -1.$$

The Matchete input is normalized before comparison by setting $hbar \to 1$ and extracting the finite ϵ^0 part term by term.

2 Canonical Normalization

The quadratic derivative sector of the Matchete result is proportional to the Fierz–Pauli kinetic term,

$$\mathcal{L}_{2,2}^{\mathrm{Matchete}} = Z_h\,\mathcal{L}_{\mathrm{FP}}, \qquad Z_h = \frac{M^2\kappa^2(1+\log(\mu^2/M^2))}{24}.$$

For the present run, the canonicalization target is chosen directly by user convention rather than inferred from the reference branch. The field rescaling therefore uses

$$\beta^2 = \frac{s_{\mathrm{kin}}}{Z_h} = -\frac{1}{Z_h},$$

so that the quadratic sector is mapped to $-\mathcal{L}_{\mathrm{FP}}$.

3 Sector Support

At $d \leq 5$, the supported reference sectors are

$$\{1, 0\},\ \{2, 0\},\ \{2, 2\},\ \{3, 0\},\ \{3, 2\},\ \{4, 0\},\ \{5, 0\}.$$

The final comparison found no unsupported leftover sectors:

$$\texttt{UnsupportedSectors} = \varnothing, \qquad \texttt{UnsupportedDifference} = 0.$$

4 Reduction Summary

4.1 Reduced Matchete Input

Sector	Input terms	IBP terms	Reduced terms	Physical coordinates
{1,0}	2	2	2	1
{2,0}	4	4	4	2
{2,2}	10	4	4	4
{3,0}	6	6	6	3
{3,2}	42	13	6	10
{4,0}	10	10	10	5
{5,0}	14	14	14	7

4.2 Reduced Reference

Sector	Input terms	IBP terms	Reduced terms	Physical coordinates
{1,0}	1	1	1	1
{1,2}	2	0	0	0
{2,0}	2	2	2	2
{2,2}	10	4	4	4
{3,0}	3	3	3	3
{3,2}	28	13	6	10
{4,0}	5	5	5	5
{5,0}	6	6	6	7

5 Solved Matching Constraints

The final saved result satisfies

$$\texttt{ExactMatchQ} = \text{True}, \qquad \texttt{HasSolution} = \text{True}.$$

The explicit solved branches in the stored result reduce to the following invariant conditions:

$$\mu^2 > 0, \tag{1}$$

$$M^2 = e^{3/2}\mu^2, \tag{2}$$

$$\Lambda_{\text{ref}} = 0, \tag{3}$$

$$\kappa_{\text{ref}}^2 = \frac{48}{M^2}. \tag{4}$$

In addition, the sign of κ_{ref} is correlated with the sign of the input κ:

$$\kappa > 0 \Rightarrow \kappa_{\mathrm{ref}} > 0, \qquad \kappa < 0 \Rightarrow \kappa_{\mathrm{ref}} < 0.$$

Equivalently, the saved `Reduce` output is

```
\[Mu]bar2 > 0 &&
((M == E^(3/4)*Sqrt[\[Mu]bar2] &&
  ((\[Kappa] > 0 && LambdaRefFinal == 0 &&
    KappaRefFinal == 4*Sqrt[3]*Sqrt[M^(-2)]) ||
   (\[Kappa] < 0 && LambdaRefFinal == 0 &&
    KappaRefFinal == -4*Sqrt[3]*Sqrt[M^(-2)]))) ||
 (M == -(E^(3/4)*Sqrt[\[Mu]bar2]) &&
  ((\[Kappa] > 0 && LambdaRefFinal == 0 &&
    KappaRefFinal == 4*Sqrt[3]*Sqrt[M^(-2)]) ||
   (\[Kappa] < 0 && LambdaRefFinal == 0 &&
    KappaRefFinal == -4*Sqrt[3]*Sqrt[M^(-2)]))))
```

6 Conclusion

After fixing the canonical-sign logic so that the kinetic target sign is an explicit user-controlled convention, the ORBIT comparison closes cleanly at $d \leq 5$ in the sign-flipped Einstein–Hilbert branch. In this final run:

- the comparison is exact in the supported sector space;
- no unsupported sectors survive;
- the cosmological constant is forced to vanish;
- the remaining parameter relations are the ones listed above.

The expensive $d \leq 6$ rerun was intentionally deferred. The current evidence therefore supports the statement:

> the canonically normalized Matchete result matches a sign-flipped Einstein–Hilbert reference through mass dimension five, once the user-chosen kinetic sign is fixed to $s_{\mathrm{kin}} = -1$.

7 Primary Artifact

The machine-readable source for this report is the saved comparison result

`orbit_fixed_eh_comparison_d5_final_result.wl`

which contains the full reduced input, reduced reference, solved constraints, and sector-by-sector comparison data.